Bullish News and Bubble Behavior in a Multiround Asset Market

EDSL shared-state pilot

EDSL replication study

October 10, 2026

Abstract

This follow-up changes two features of the generic-prompt pilot: all 12 traders receive explicit behavioral trading instructions, and six receive a strongly bullish private analyst forecast after period 15. Four participants are momentum buyers, four are bubble riders, and four are profit-taking sellers. The one completed 30-period session produced 22 shares traded. Prices rose from \$15.25 in period 1 to \$17.90 in period 4, then trading paused until periods 27–29, when prices were \$17.75–\$18.00. The prompt design elicited active, two-sided orders and a brief early price run-up, but the long no-trade interval means this is not evidence of a sustained bubble or a causal news effect. This note documents the treatment, shared-state machine, workflow, and results.

1 Design

The 30-period call market uses 12 traders, each starting with \$100 cash and four shares. Cash earns 5% interest; per-share dividends are \$0.40 or \$1.00 with equal probability; the exchange clears a uniform marginal-midpoint price; and remaining shares are redeemed for \$14 after period 30. The shared public tape reports prices, volume, and dividends. Traders see their account, personal history, and the public tape.

All participants receive one of three behavioral prompts: four momentum buyers, four bubble riders, and four profit-taking sellers. Opening instructions name concrete limit-price ranges and quantities. This is intended to create buyers and sellers from the opening periods rather than relying on spontaneous coordination by identical generic traders.

After period 15 settles, two traders in each behavioral group receive a private bullish analyst update. The report forecasts that resale prices could reach \$20 by period 25 because institutional demand is unusually strong. It is described as a forecast rather than a guaranteed payoff; dividends and actual terminal redemption remain unchanged. Seed 141027 fixes the behavior composition and stratified assignment. Recipients: trader-00, trader-01 (momentum); trader-04, trader-05 (bubble riders); trader-09, trader-11 (profit takers).

2 Results

Measure	Pilot result
Trader decisions	360
Settlements	30 / 30
Private update deliveries	6
Completed workflow items	396 / 396
Shares traded	22
Periods with trade	7 / 30
Peak observed transaction price	$18.00
Terminal shares	0 for every trader
Terminal cash range	$642.76–$717.52

Transactions occurred in periods 1–4 and 27–29. Prices in the opening sequence were $15.25, $16.63, $17.25, and $17.90. The late sequence was $18.00, $17.75, and $17.75. No transactions occurred in periods 5–26. Across all decisions, participants submitted 181 buy orders and 39 sell orders. Momentum traders submitted 78 buys; bubble riders submitted 103 buys; profit takers submitted 15 sells. Many orders did not fill, so these counts describe intended orders rather than completed trades.

The mean current-price forecast for update recipients was $17.77 in period 15, $17.93 in period 16, and $18.17 in period 25. For traders without the update, the corresponding means were $17.77, $17.77, and $17.90. This small descriptive gap is consistent with some response to the news, but it is not a causal estimate. Traders share a market, and there is only one session.

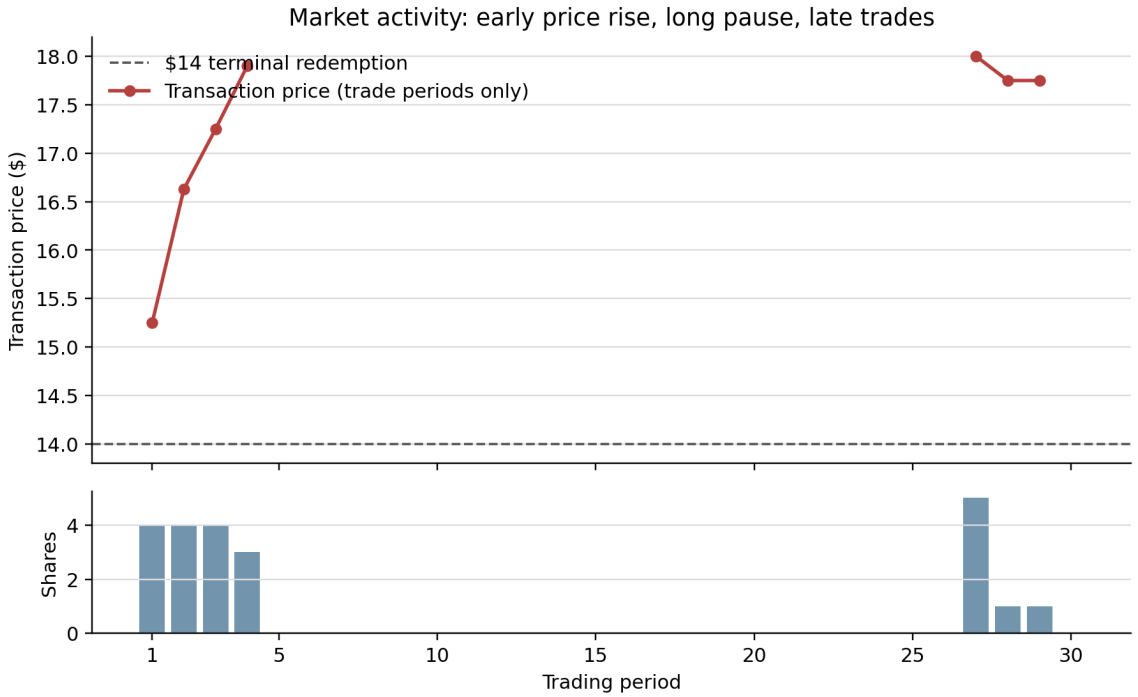


Figure 1: Observed transaction prices and traded volume. The $14 line is the terminal redemption amount. Missing price points indicate periods with no transaction.

The prompts produced a brief opening run-up above the terminal redemption amount, followed by 22 inactive periods and three late trading periods. The result is a useful pilot for a bubble-inducing instruction set, not a sustained bubble. The next evidence should come from multiple independently

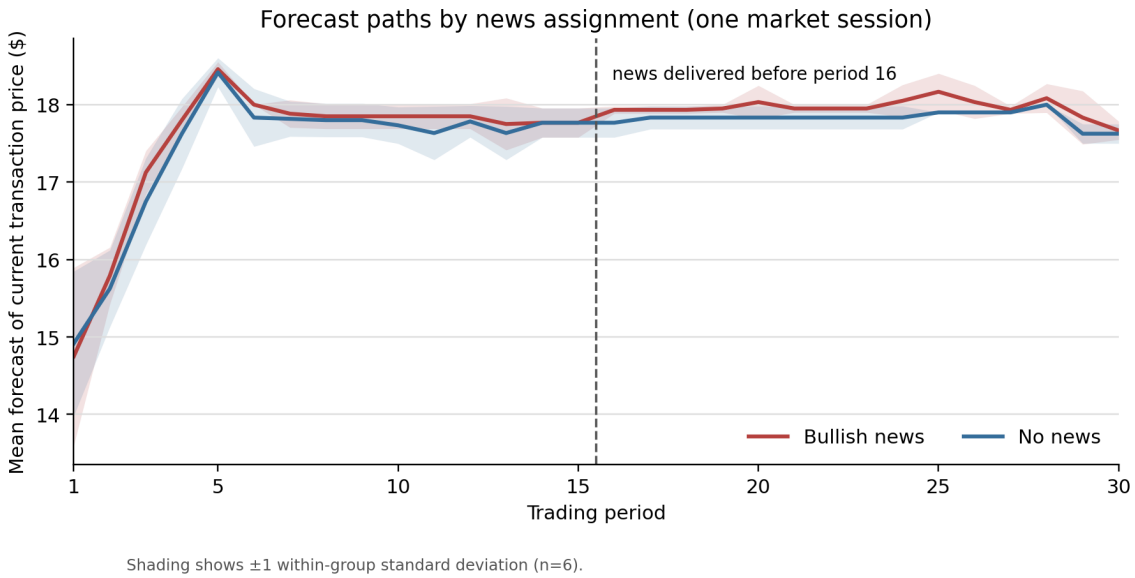


Figure 2: Mean current-price forecasts by update assignment. Shading is one within-group standard deviation ($n = 6$), not a confidence interval. The update was delivered before period 16.

seeded markets, preferably with enough sessions to estimate variation across market composition and treatment assignment.

3 Shared-state machine

The experiment starts from EDSL's native 'call_market' machine, which defines accounts, orders, order history, tape, period, last price, settlement, interest, dividends, and redemption. It adds a private update map and one command. The machine's public view includes the trader's account and shared tape; the added projection selects only the update for the current trader.

```
machine = call_market(TRADERS, rules=RULES, seed=SEED)
machine.fields["private_updates"] = state_field(StateType.map(), {})
machine.commands["publish_update"] = Command(
    inputs={"recipient": StateType.text(), "message": StateType.text()},
    effects=(put("private_updates", arg("recipient"), arg("message")),),
)
machine.view["your_update"] = field("private_updates").get(current("name"))
machine.validate()
```

The private map itself is not present in the trader-facing view. The DSL lookup uses the current participant's name to select one entry and yields an update only to that participant. Everyone else gets an empty result rendered as "No private analyst update has been issued to you." Each publication is a separate workflow write, so recipient, message, and event order are persisted and auditable.

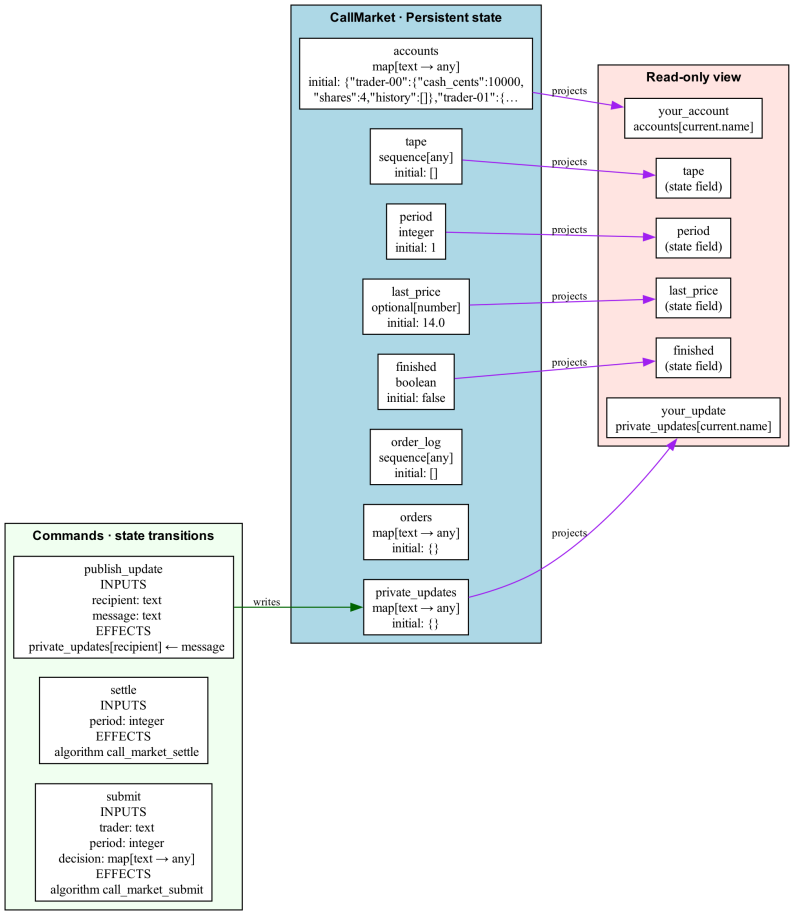


Figure 3: Detailed diagram generated by EDSL's 'Machine.show_diagram' method.

4 Round execution and midpoint update

Each period consists of a trader order step followed by exchange settlement. Trader work reads the machine and writes each structured decision through ‘submit‘; settlement calls the native deterministic market algorithm. After period 15 settlement, six scripted exchange items publish the updates in a chain. Period 16 trader work waits for all six writes, so the message is absent from earlier views and available in the intended recipients’ next decisions.

```
orders = workflow.step(
    f"orders-{period}", Survey([decision]),
    assigned_to=role("trader"), after=previous,
    reads=(market.read(),),
    writes=(market.submit(
        trader=current.agent.name, period=period,
        decision=decision.answer,
    ),),
)
settled = workflow.step(
    f"settle-{period}", settlement_survey,
    assigned_to=role("exchange"), after=orders,
    writes=(market.settle(period=period),),
)
previous = settled
if period == 15:
    for recipient in recipients:
        previous = workflow.step(
            f"publish-private-update-{recipient}", update_survey,
            assigned_to=role("exchange"), after=previous,
            writes=(market.publish_update(
                recipient=recipient, message=UPDATE,
            ),),
        )
```

The six update tasks use a scripted exchange answer, so they create no model calls. The trader behavior is stored in each agent’s traits and supplied as behavioral instructions at every decision period. Treatment status is a separate trait and does not replace the behavior persona.

5 Run command

Preparation writes the serializable experiment and machine without inference:

```
python3 edsl_replication/studies/asset_market_bullish_bubble/experiment.py \
  --output edsl_replication/studies/asset_market_bullish_bubble/pilot_01 \
  --prepare-only
```

The pilot was launched locally through EDSL using the configured provider credentials, with ‘–prepare-only‘ omitted. EDSL wrote its workflow database, shared-state SQLite log, model calls, and job artifacts under ‘pilot_01/‘.

6 Limits

This is one market with a mixed treatment and one common tape. Information can spread through observed behavior and transaction prices, so traders are not independent experimental units. The strongly positive analyst forecast is belief-relevant but does not change realized dividend or redemption

rules. The 22 shares traded and short opening price run-up motivate replications; they do not establish a bubble or the causal effect of news or behavioral prompts.